Moaz Mamdouh Ayoub

AI Engineer

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Portfolio: https://moaz-mamdouh207.github.io/

SUMMARY

AI Engineer specializing in Retrieval-Augmented Generation (RAG) systems and agentic AI applications. With a strong foundation in machine learning and software engineering. Strong academic performance in engineering with consistent top ranking.

OBJECTIVE

Seeking an AI Engineering internship where I can apply and expand my skills in RAG systems, agentic AI, and LLM application development, while contributing to building scalable and production-ready AI solutions in a real-world engineering environment.

EDUCATION

Bachelor of Science in Mechatronics Engineering **Oct 2022 - July 2027 (Expected)**

- 6 October University, Giza, Egypt
- GPA: 3.87/4 - Excellent
- Consistently ranked 1st in class throughout 4 academic years

PROJECTS

UNI-RAG (Multimodal Agentic Retrieval-Augmented Generation Platform)

- Developed a multimodal Agentic RAG system enabling knowledge ingestion from PDF, PowerPoint, and Word documents and supporting text, image, and PDF-based queries.
- Built scalable backend APIs using FastAPI and implemented asynchronous document processing with Celery and Redis.
- Integrated vLLM for custom document parsers during ingestion to handle scanned documents.
- Designed semantic retrieval pipelines using PostgreSQL, pgvector, Qdrant achieving **93%** recall (evaluated using a custom dataset and RAGAS framework)
- Full agentic pipeline with calculator tool to avoid mathematical errors in engineering textbooks.
- Citations in the answer showing the exact pages used from the documents and why they are used.
- Developed a React-based GUI.

Tech Stack: Python, FastAPI, LangChain, PostgreSQL, pgvector, SQLAlchemy, Redis, Celery, React, RAGAS, LangFuse

Demo Video : https://youtu.be/C0G3PML704A
Source Code : https://github.com/moaz-mamdouh207/uni_rag

Voice Controlled Wheelchair

- Developed a voice-controlled wheelchair system using CNN-based audio command recognition
- Implemented and deployed ML models using TensorFlow Lite + C++
- Co-authored a research paper comparing traditional machine learning methods with deep learning approaches for audio recognition tasks

Tech Stack: TensorFlow Lite

Source Code: https://github.com/moaz-mamdouh207/Voice-Controlled-Wheelchair

SKILLS

- **Technical Skills:**
  - Programming Languages: Python
  - Frameworks: FastAPI
  - AI / LLM Frameworks: LangChain (RAG pipelines, agent-based systems)
  - Databases: PostgreSQL, vector databases (Qdrant, pgvector)
  - Task Processing & Caching: Celery, Redis
  - Version Control: Git
  - Evaluation Tools: Ragas
  - Machine Learning: Machine learning techniques, deep learning, natural language processing (NLP)
- **Soft skills:**

  Problem-solving mindset, Self-motivation, Adaptability, Critical Thinking and Continuous learning

- **Languages:**

  Arabic (Native) | English (very good)

COURSES

McKinsey Forward Program **Oct 2025 - Dec 2025**

Provided by: McKinsey & Company

- Completed the McKinsey Forward Program, developing structured problem-solving, analytical thinking, communication, and collaboration skills.

AI & Data Science - Microsoft Machine Learning Engineer **June 2025 - Dec 2025**

Provided by: DEPI | Ministry of Communications and Information Technology

- Python-based AI development covering data preprocessing, visualization, and machine learning
- Deep learning, NLP (attention mechanisms), and computer vision with transfer learning
- MLOps and deployment using MLflow, Hugging Face, and Microsoft Azure AI services